

Climate 5000 Series - Gen 4

24K BTU Single Zone Regular System

Ducted IDU

Engineering
 Submittal
 Sheet



BOSCH

Overview and Project Information



Project name	
Location	
Architect	
Engineer	
General contractor	
Reference #	

24K Btu Single Zone Regular System - Ducted IDU

Indoor part number	8-733-962-712
Indoor model number	BMS500-AAU024-1AHDXD
Outdoor part number	8-733-962-687
Outdoor model number	BMS500-AAS024-1CSXRD

Efficiency Data		AHRI 210/240 2023
SEER2		19.0
EER2		12.5
HSPF2 Region IV		11.0
HSPF2 Region V		8.5
COP(47F)		3.64
COP (5F)		2.30

Performance Data		AHRI 210/240 2023
Cooling	Capacity Range	8500~28300
	Rated Capacity	24000
	Current	8.3
	Input Power	1920
Heating	Capacity Range	11800~35100
	Rated Capacity (47F)	25000
	Rated Capacity (5F)	19000
	Current	8.7
	Input Power	2010

Electrical Data	
Power supply (V / Ø / Hz)	208-230 / 1 / 60
Min. circuit ampacity indoor/outdoor (A)	3.0 / 24.0
Max. fuse indoor/outdoor (A)	15.0 / 25.0

Compressor Data	
Type:	Rotary
Refrigerant oil /oil charge (ml)	VG74 / 620
Input (W)	805 / 2045
Rated current (RLA/A)	4.15 / 9.30

Piping	
Refrigerant precharge (oz/kg)	70.55 / 2.00
Length for precharge (ft/m)	24.6 / 7.5
Liquid side (in/mm)	3/8" / Ø9.52
Gas side (in/mm)	5/8" / Ø15.9
Additional Charge of Refrigerant (oz/ft)/(g/m)	0.32 / 30
Max. refrigerant pipe length (ft/m)	164 / 50
Max. difference in height (ft/m)	82 / 25

Operation Range

ODU	Cooling (°F / °C)	-13 ~ 122 / -25 ~ 50
	Heating (°F / °C)	-13 ~ 75 / -25 ~ 24
IDU	Cooling (°F / °C)	61 ~ 90 / 16 ~ 32
	Heating (°F / °C)	32 ~ 86 / 0 ~ 30

System Data

Refrigerant type		R454B
Design pressure (PSIG)		550 / 340
Dehumidification (pts/hr)		4.52
IDU	Fan motor model	ZKFN-165-10-1L
	Fan motor qty (ea)	1
	Fan motor input (W)	185.0
	Fan speed, H / M / L (RPM)	920 / 835 / 750
	Air flow, Turbo / H / M / L / Si (CFM)	853.5 / 706.3 / 618.0 / 500.3 / N/A
	Noise level, Turbo / H / M / L / Si (dB(A))	43 / 40.0 / 38 / 35.5 / 29.5
	Net weight (lbs / kg)	80.47 / 36.5
	Gross weight (lbs / kg)	91.93 / 41.7
ODU	ESP range (Pa)	0-200
	Drain pipe outside diameter in. (mm)	OD 0.98 (25)
	Fan motor model	ZKFN-120-10-1L
	Fan motor qty (ea)	1
	Fan motor input (W)	170
	Fan motor speed, H / / L (RPM)	810 / 700 / 450
	Air flow, Max (CFM)	2236.7
	Noise level, Max (dB(A))	60.0
	Net weight (lbs / kg)	122.13 / 55.4
	Gross weight (lbs / kg)	132.28 / 60.0

Accessories

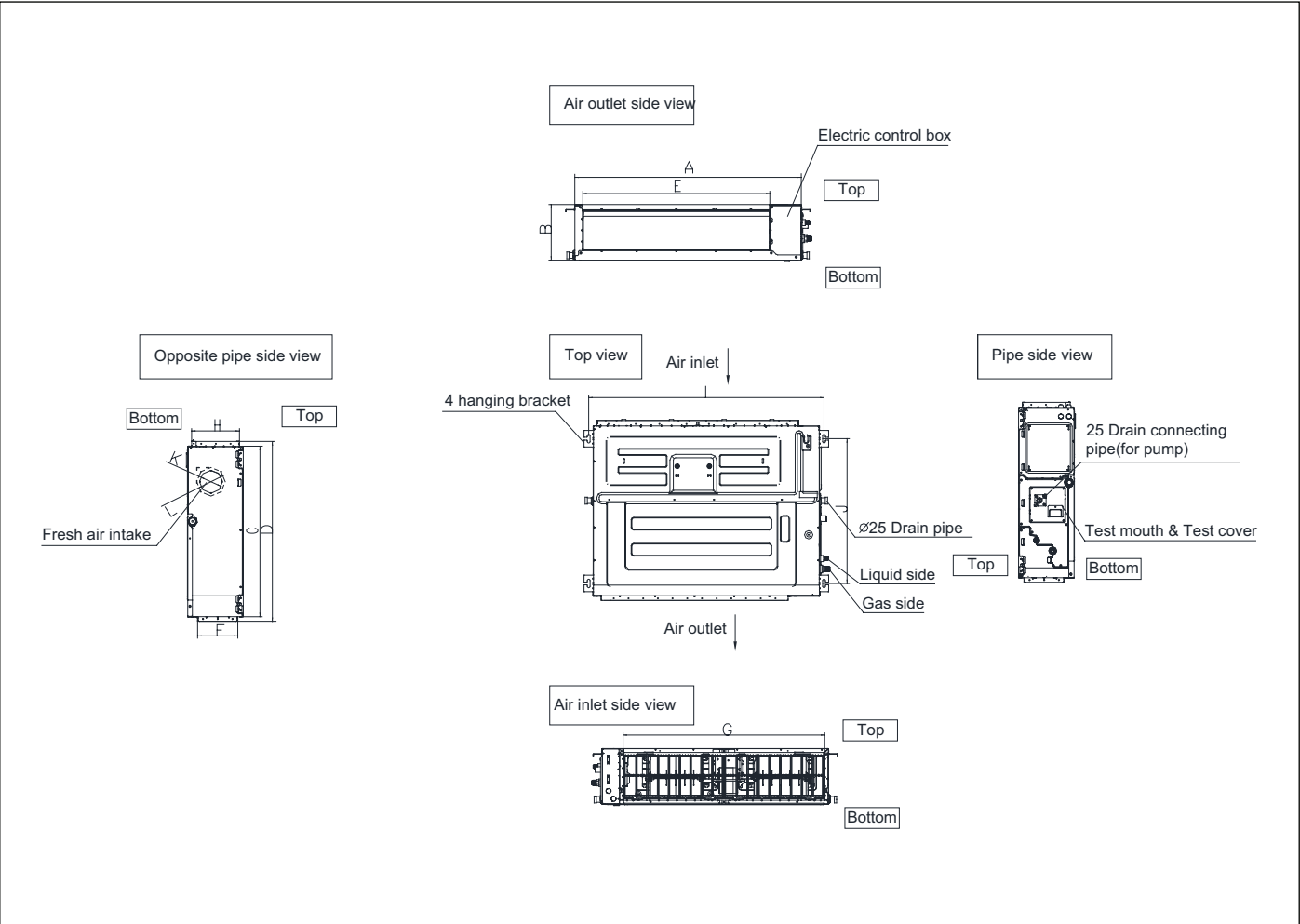
BMS-WT2-XXD (8733962707)	Wired Wall Thermostat
BMS-DRY-XXD (8733969263)	24V Interface
BMS-PK3-7X1 (8733951015)	25ft Lineset

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Indoor Unit Dimensions and Connections

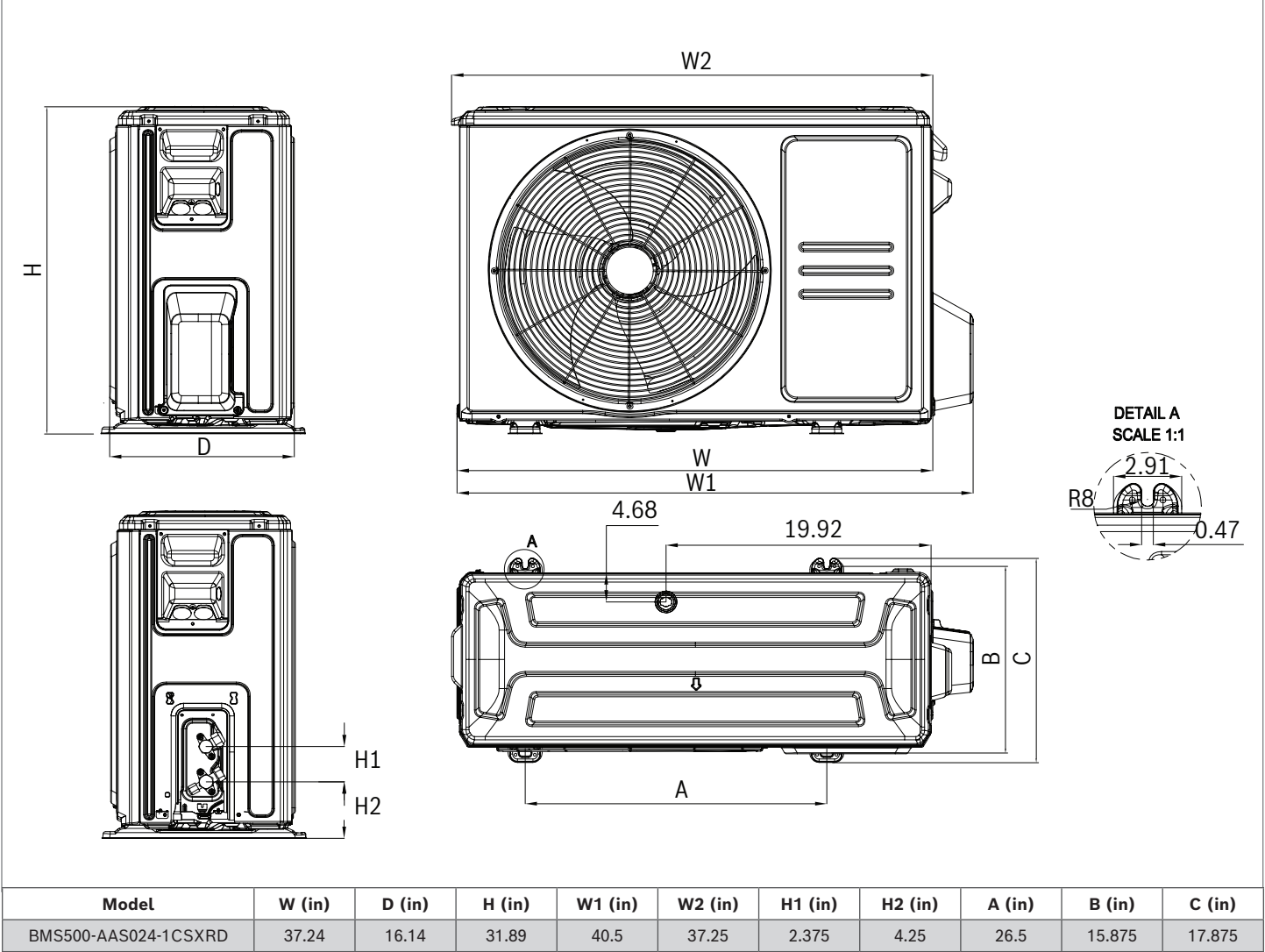


Model	Outline Dimension (in)				Air Outlet Opening Size (in)		Air Return Opening Size (in)		Size of Mounted Lug (in)		Fresh Air Intake Opening Size (in)	
	A	B	C	D	E	F	G	H	I	J	K	L
BMS500-AAU024-1AHDXD	39.37	9.6	29.5	31.3	32.6	7.0	35.1	8.3	40.9	25.2	3.9	5.0

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Outdoor Unit Dimensions and Connections



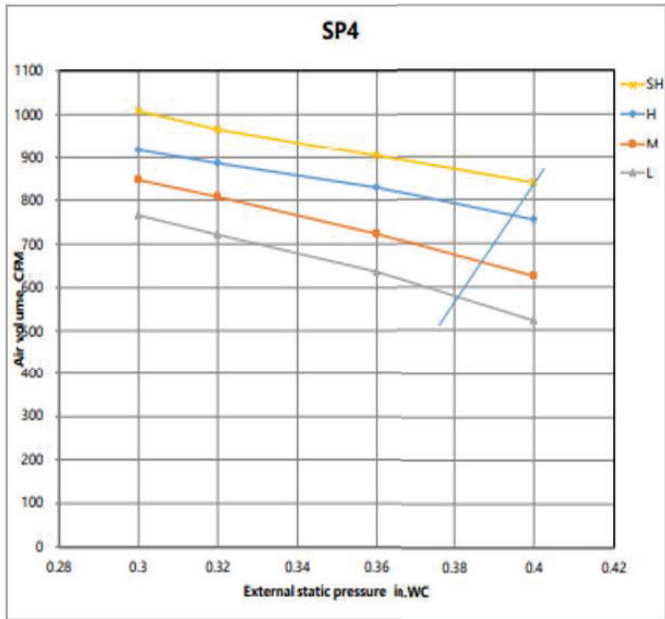
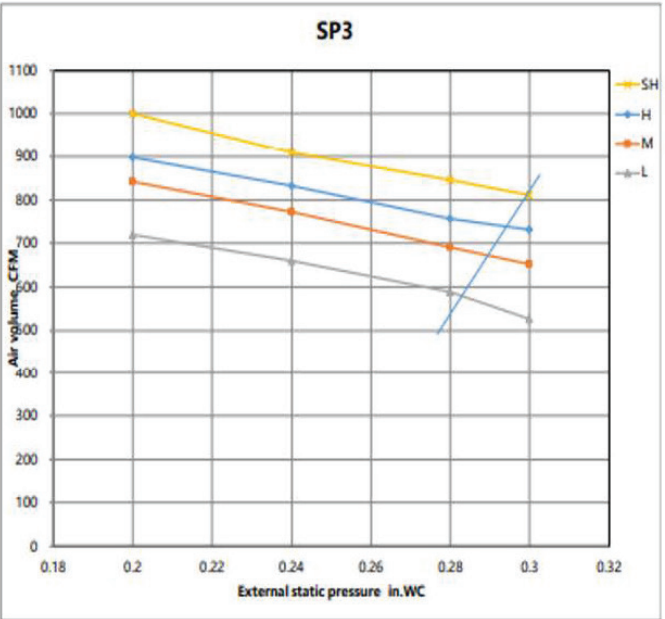
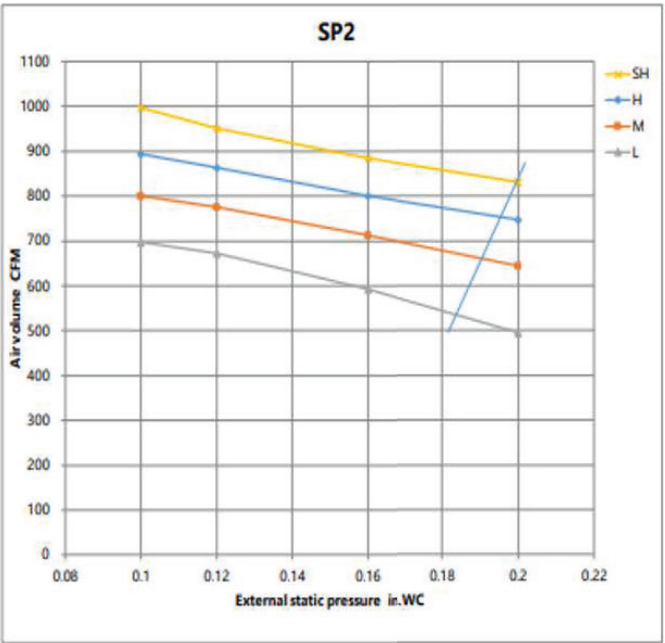
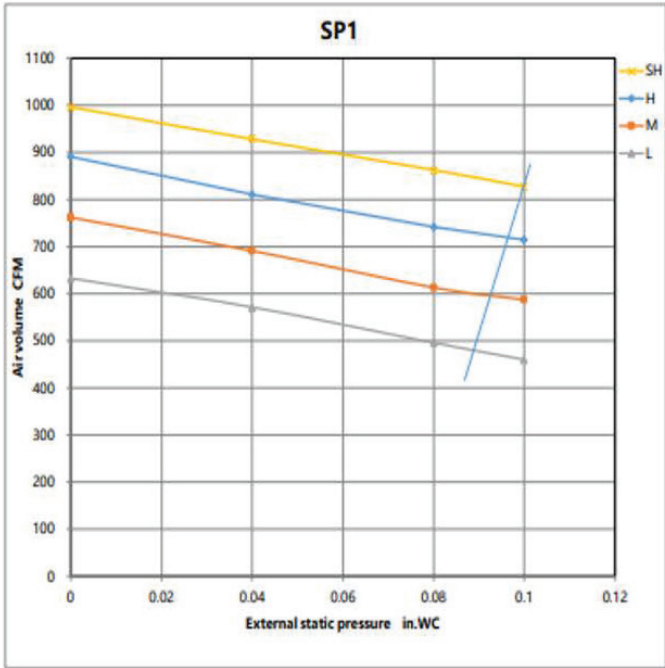
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24K Btu Single Zone Regular System - Ducted IDU

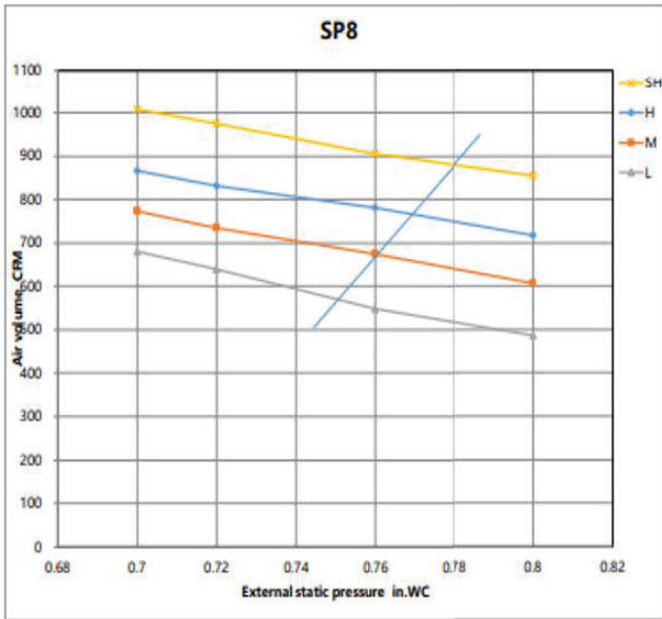
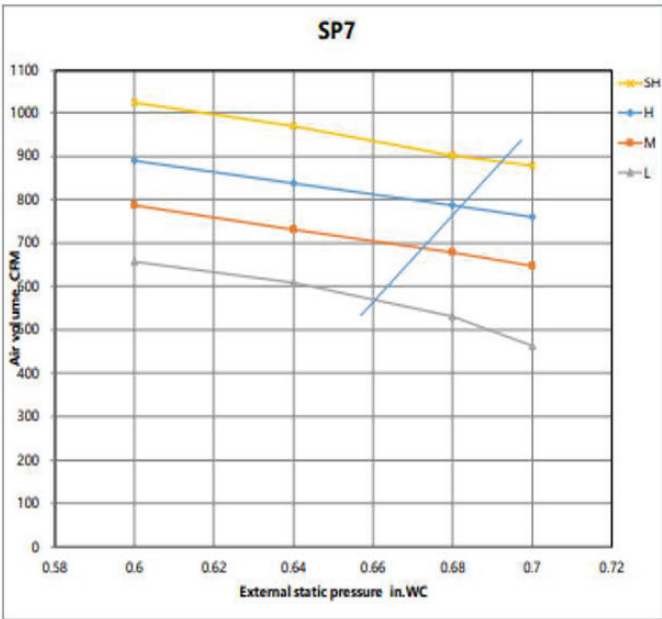
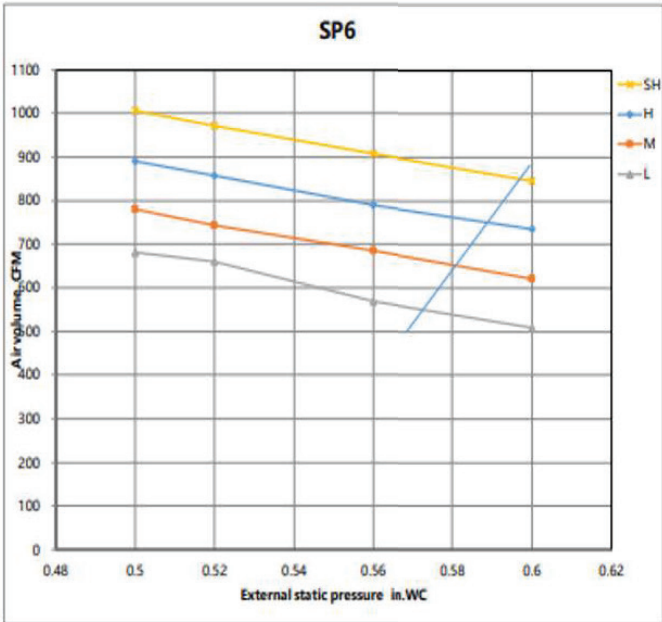
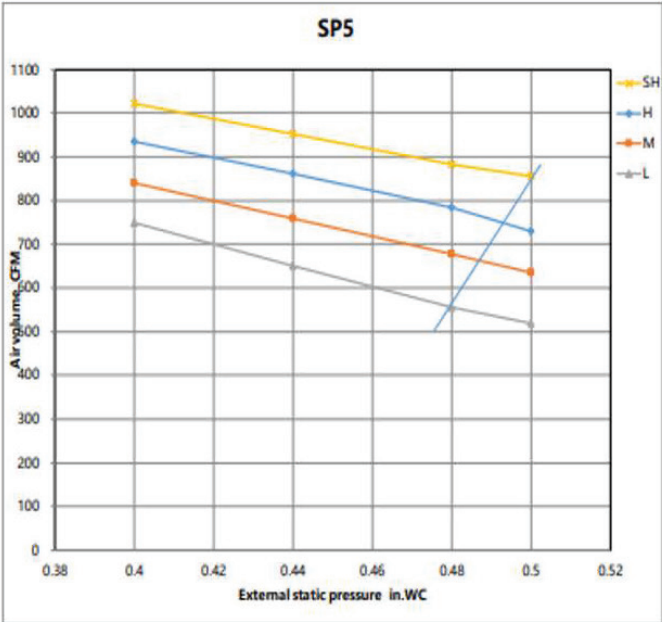


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Air Volume

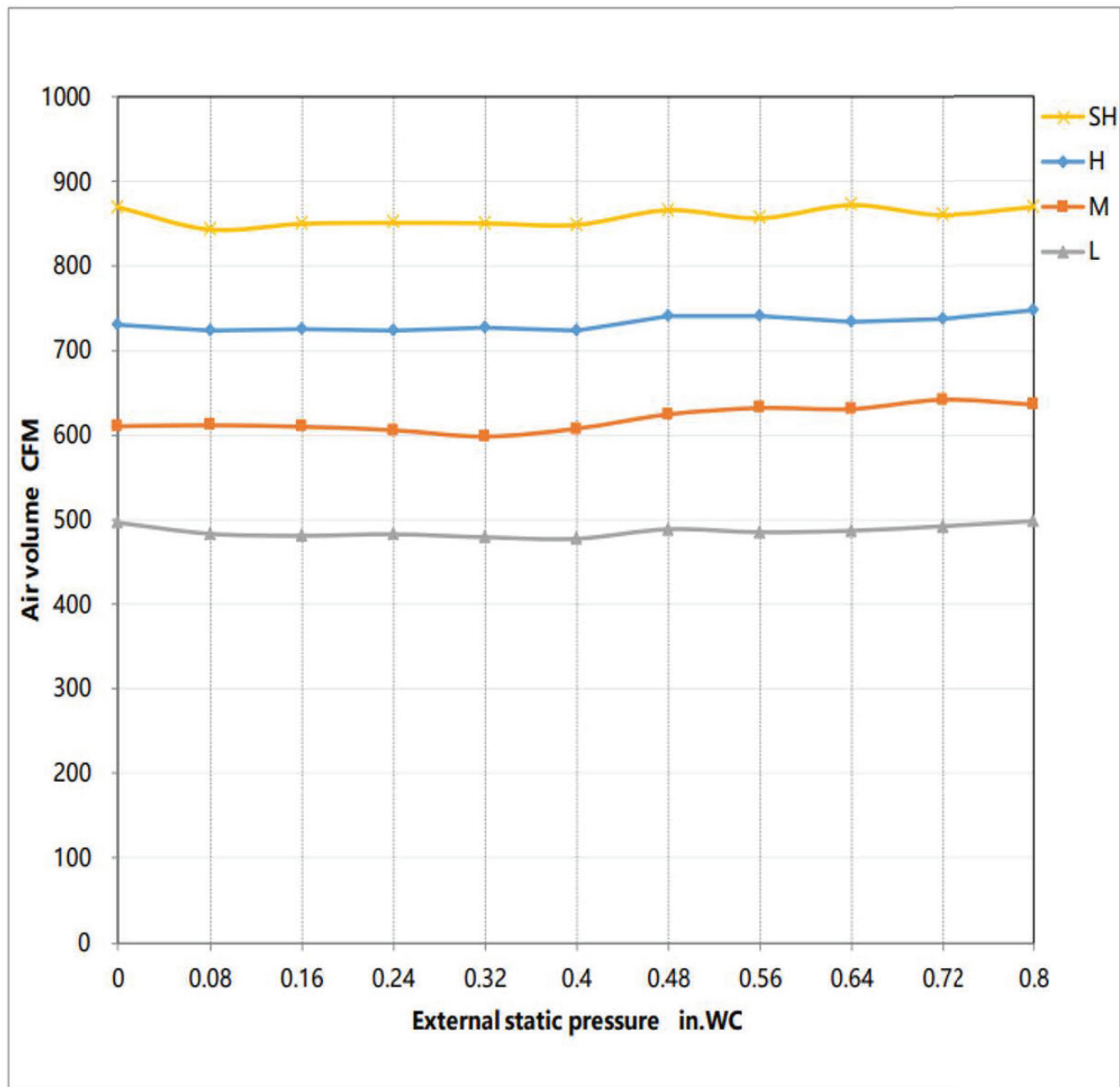


Air Volume



Air Volume

Constant air volume



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Cooling Performance - Indoor Dry Bulb Temperature at 1235 CFM																					
ODU DB(°F)	IDU WB (°F)	60.8°F(16.0°C)					64.4°F(18.0°C)					67°F(19.4°C)					71.6°F(22.0°C)				
	IDU DB (°F)	73.4°F (23.0°C)	77.0°F (25.0°C)	80.0°F (26.7°C)	80.6°F (27.0°C)	84.2°F (29.0°C)	73.4°F (23.0°C)	77.0°F (25.0°C)	80.0°F (26.7°C)	80.6°F (27.0°C)	84.2°F (29.0°C)	73.4°F (23.0°C)	77.0°F (25.0°C)	80.0°F (26.7°C)	80.6°F (27.0°C)	84.2°F (29.0°C)	73.4°F (23.0°C)	77.0°F (25.0°C)	80.0°F (26.7°C)	80.6°F (27.0°C)	84.2°F (29.0°C)
-13°F (-25.0°C)	TC	20239	20267	20288	20289	20317	21459	21484	21506	21510	21535	22741	22771	22796	22800	22826	24111	24138	24164	24168	24198
	S/T	0.48	0.56	0.64	0.64	0.73	0.51	0.60	0.68	0.68	0.77	0.54	0.63	0.72	0.72	0.82	0.57	0.67	0.76	0.76	0.87
	PI	1.00	1.01	1.01	1.01	1.01	1.07	1.07	1.07	1.07	1.07	1.13	1.13	1.13	1.13	1.13	1.20	1.20	1.20	1.20	1.20
-4°F (-20.0°C)	TC	21796	21826	21849	21850	21879	23110	23137	23160	23165	23192	24490	24522	24550	24554	24582	25966	25994	26023	26027	26059
	S/T	0.48	0.57	0.64	0.64	0.73	0.51	0.60	0.68	0.68	0.78	0.54	0.64	0.72	0.72	0.82	0.57	0.67	0.77	0.77	0.87
	PI	1.08	1.08	1.08	1.08	1.09	1.15	1.15	1.15	1.15	1.15	1.22	1.22	1.22	1.22	1.22	1.29	1.29	1.29	1.29	1.29
0°F (-17.8°C)	TC	22488	22518	22543	22544	22574	23843	23872	23895	23900	23928	25268	25301	25329	25334	25362	26790	26819	26849	26853	26886
	S/T	0.48	0.57	0.64	0.64	0.73	0.51	0.60	0.68	0.68	0.78	0.54	0.64	0.72	0.72	0.83	0.58	0.68	0.77	0.77	0.88
	PI	1.12	1.12	1.12	1.12	1.12	1.18	1.19	1.19	1.19	1.19	1.25	1.26	1.26	1.26	1.26	1.33	1.33	1.33	1.33	1.33
5°F (-15.0°C)	TC	23353	23384	23410	23411	23442	24760	24790	24814	24819	24849	26240	26274	26303	26308	26337	27821	27851	27881	27886	27920
	S/T	0.49	0.57	0.65	0.65	0.74	0.51	0.60	0.69	0.69	0.78	0.55	0.64	0.73	0.73	0.83	0.58	0.68	0.77	0.77	0.88
	PI	1.16	1.16	1.16	1.16	1.16	1.23	1.23	1.23	1.23	1.23	1.30	1.30	1.31	1.31	1.31	1.38	1.38	1.38	1.38	1.39
14°F (-10.0°C)	TC	21783	21812	21836	21837	21866	23096	23123	23146	23151	23178	24476	24508	24535	24539	24567	25950	25979	26007	26011	26043
	S/T	0.49	0.57	0.65	0.65	0.74	0.52	0.61	0.69	0.69	0.78	0.55	0.64	0.73	0.73	0.83	0.58	0.68	0.77	0.77	0.88
	PI	1.14	1.14	1.14	1.14	1.14	1.21	1.21	1.21	1.21	1.21	1.28	1.28	1.28	1.28	1.28	1.36	1.36	1.36	1.36	1.36
17°F (-8.3°C)	TC	21260	21288	21311	21312	21341	22541	22568	22590	22594	22621	23888	23919	23945	23950	23977	25327	25355	25382	25386	25418
	S/T	0.49	0.57	0.65	0.65	0.74	0.52	0.61	0.69	0.69	0.79	0.55	0.64	0.73	0.73	0.83	0.58	0.68	0.78	0.78	0.88
	PI	1.13	1.13	1.13	1.13	1.13	1.20	1.20	1.20	1.20	1.20	1.27	1.27	1.27	1.27	1.27	1.35	1.35	1.35	1.35	1.35
23°F (-5.0°C)	TC	21920	21950	21973	21974	22004	23241	23269	23292	23296	23324	24630	24662	24689	24694	24721	26114	26142	26171	26175	26207
	S/T	0.49	0.58	0.65	0.65	0.75	0.52	0.61	0.69	0.69	0.79	0.55	0.65	0.74	0.74	0.84	0.58	0.69	0.78	0.78	0.89
	PI	1.09	1.09	1.09	1.09	1.09	1.16	1.16	1.16	1.16	1.16	1.22	1.23	1.23	1.23	1.30	1.30	1.30	1.30	1.30	1.30
35°F (1.7°C)	TC	23241	23272	23297	23298	23330	24642	24671	24695	24700	24729	26114	26148	26177	26182	26211	27687	27717	27747	27752	27786
	S/T	0.49	0.58	0.66	0.66	0.75	0.52	0.61	0.70	0.70	0.79	0.55	0.65	0.74	0.74	0.84	0.59	0.69	0.78	0.78	0.89
	PI	1.01	1.01	1.01	1.01	1.01	1.07	1.07	1.07	1.07	1.07	1.13	1.13	1.14	1.14	1.14	1.20	1.20	1.20	1.20	1.20
41°F (5.0°C)	TC	23333	23365	23390	23391	23423	24740	24769	24793	24798	24828	26218	26252	26281	26286	26315	27797	27828	27858	27862	27897
	S/T	0.49	0.58	0.66	0.66	0.75	0.52	0.61	0.70	0.70	0.80	0.56	0.65	0.74	0.74	0.84	0.59	0.69	0.78	0.78	0.89
	PI	1.07	1.07	1.07	1.07	1.07	1.13	1.13	1.13	1.13	1.14	1.20	1.20	1.20	1.20	1.20	1.27	1.27	1.27	1.27	1.28
47°F (8.3°C)	TC	23426	23457	23483	23484	23515	24838	24867	24892	24896	24926	26321	26356	26385	26390	26419	27907	27938	27968	27972	28007
	S/T	0.50	0.58	0.66	0.66	0.75	0.53	0.62	0.70	0.70	0.80	0.56	0.65	0.74	0.74	0.85	0.59	0.69	0.79	0.79	0.90
	PI	1.13	1.13	1.13	1.13	1.13	1.19	1.20	1.20	1.20	1.20	1.26	1.27	1.27	1.27	1.27	1.34	1.34	1.34	1.34	1.35
50°F (10.0°C)	TC	23731	23763	23789	23790	23822	25161	25191	25216	25221	25251	26665	26699	26729	26734	26764	28271	28302	28333	28337	28372
	S/T	0.50	0.58	0.66	0.66	0.76	0.53	0.62	0.70	0.70	0.80	0.56	0.66	0.75	0.75	0.85	0.59	0.70	0.79	0.79	0.90
	PI	1.15	1.15	1.15	1.15	1.16	1.22	1.22	1.22	1.22	1.22	1.29	1.29	1.30	1.30	1.30	1.37	1.37	1.37	1.37	1.38
59°F (15.0°C)	TC	24647	24681	24707	24708	24742	26133	26164	26190	26195	26226	27694	27730	27761	27766	27797	29363	29395	29427	29431	29468
	S/T	0.50	0.59	0.67	0.67	0.76	0.53	0.62	0.71	0.71	0.81	0.56	0.66	0.75	0.75	0.86	0.60	0.70	0.80	0.80	0.91
	PI	1.23	1.23	1.23	1.23	1.23	1.30	1.30	1.30	1.30	1.30	1.38	1.38	1.38	1.38	1.46	1.46	1.46	1.46	1.47	
68°F (20.0°C)	TC	25564	25598	25626	25627	25661	27104	27136	27163	27168	27201	28724	28761	28793	28830	30454	30487	30520	30525	30563	
	S/T	0.51	0.60	0.69	0.69	0.78	0.54	0.64	0.73	0.73	0.83	0.58	0.68	0.77	0.77	0.88	0.61	0.72	0.82	0.82	0.93
	PI	1.30	1.30	1.30	1.30	1.31	1.38	1.38	1.38	1.38	1.38	1.46	1.46	1.47	1.47	1.55	1.55	1.55	1.55	1.56	
77°F (25.0°C)	TC	26480	26515	26544	26545	26581	28076	28109	28137	28142	28176	29753	29792	29825	29830	29864	31546	31580	31614	31619	31659
	S/T	0.52	0.61	0.69	0.69	0.79	0.55	0.65	0.74	0.74	0.84	0.59	0.69	0.78	0.78	0.89	0.62	0.73	0.83	0.83	0.94
	PI	1.38	1.38	1.38	1.38	1.38	1.46	1.46	1.46	1.46	1.46	1.55	1.55	1.55	1.55	1.55	1.64	1.64	1.64	1.64	1.65
82°F (27.8°C)	TC	26570	26606	26634	26635	26671	28171	28204	28232	28238	28271	29854	29893	29926	29932	29965	31653	31687	31722	31727	31766
	S/T	0.53	0.62	0.70	0.70	0.80	0.56	0.66	0.75	0.75	0.85	0.59	0.70	0.79	0.79	0.90	0.63	0.74	0.84	0.84	0.95
	PI	1.53	1.53	1.54	1.54	1.54	1.62	1.63	1.63	1.63	1.63	1.72	1.72	1.73	1.73	1.73	1.83	1.83	1.83	1.83	1.83
86°F (30.0°C)	TC	26113	26148	26177	26178	26213	27687	27720	27747	27753	27786	29341	29379	29412	29417	29450	31109	31143	31177	31182	31220
	S/T	0.53	0.63	0.71	0.71	0.81	0.57	0.66	0.75	0.75	0.86	0.60	0.70	0.80	0.80	0.91	0.64	0.75	0.85	0.85	0.97
	PI	1.80	1.80	1.81	1.81	1.81	1.91	1.91	1.91	1.91	1.92	2.02	2.03	2.03	2.03	2.03	2.15	2.15	2.15	2.15	2.15
95°F (35.0°C)	TC	25086	25120	25147	25148	25182	26598	26629	26655	26661	26692	28187	28223	28255	28260	28292	29885	29917	29950	29955	29992
	S/T	0.53	0.63	0.71	0.71	0.81	0.57	0.66	0.75	0.75	0.86	0.60	0.70	0.80	0.80	0.91	0.64	0.75	0.85	0.85	0.97
	PI	2.41	2.41	2.41	2.41	2.41	2.55	2.55	2.55	2.56	2.56	2.70	2.71	2.71	2.71	2.71	2.87	2.87	2.87	2.87	2.88
104°F (40.0°C)	TC	21329	21358	21381	21382	21411	22615	22642	22664	22669	22695	23966	23997	24024	24028	24055	25410	25438	25465	25469	25501
	S/T	0.55	0.64	0.73	0.73	0.83	0.58	0.68	0.77	0.77	0.88	0.62	0.72	0.82	0.82	0.93	0.65	0.76	0.87	0.87	0.99
	PI	2.29	2.																		

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Heating Performance - Indoor Dry Bulb Temperature at 1235 CFM										
Outdoor DB(°F)	TCI Total Heating Capacity(BTU/hr)					PI Power Input(kW)				
	Indoor Conditions(DB °F)					Indoor Conditions(DB °F)				
	60.8°F (16.0°C)	68°F (20.0°C)	70°F (21.1°C)	71.6°F (22.0°C)	75.2°F (24.0°C)	60.8°F (16.0°C)	68°F (20.0°C)	70°F (21.1°C)	71.6°F (22.0°C)	75.2°F (24.0°C)
-13°F(-25.0°C)	12684	12431	12369	12319	12208	2.10	2.11	2.12	2.12	2.13
-4°F(-20.0°C)	15484	15174	15098	15038	14902	2.22	2.24	2.25	2.25	2.26
0°F(-17.8°C)	17269	16923	16839	16771	16621	2.27	2.29	2.29	2.30	2.31
5°F(-15.0°C)	19500	19110	19015	18939	18768	2.33	2.35	2.36	2.36	2.37
14°F(-10.0°C)	21445	21016	20911	20828	20640	2.42	2.43	2.44	2.44	2.45
17°F(-8.3°C)	22093	21652	21543	21457	21264	2.44	2.46	2.47	2.47	2.48
22°F(-5.6°C)	23492	23022	22907	22816	22610	2.55	2.57	2.58	2.58	2.59
27°F(-2.8°C)	24891	24393	24271	24174	23956	2.66	2.69	2.69	2.70	2.71
32°F(-0.0°C)	26290	25764	25635	25532	25303	2.77	2.80	2.80	2.81	2.82
35°F(1.7°C)	27129	26586	26453	26347	26110	2.84	2.86	2.87	2.87	2.89
37°F(2.8°C)	28598	28026	27886	27775	27525	2.86	2.89	2.89	2.90	2.91
42°F(5.6°C)	32272	31627	31469	31343	31061	2.92	2.94	2.95	2.95	2.97
44.6°F(7.0°C)	33742	33067	32902	32770	32475	2.94	2.96	2.97	2.98	2.99
47°F(8.3°C)	35947	35228	35051	34911	34597	2.97	3.00	3.00	3.01	3.02
52°F(11.1°C)	36540	35810	35631	35488	35169	2.79	2.81	2.82	2.83	2.84
57°F(13.9°C)	37134	36392	36210	36065	35740	2.61	2.63	2.64	2.64	2.65
59°F(15.0°C)	37372	36624	36441	36295	35969	2.54	2.56	2.56	2.57	2.58
62°F(16.7°C)	37728	36973	36789	36641	36312	2.43	2.45	2.45	2.46	2.47
64.4°F(18.0°C)	37966	37206	37020	36872	36540	2.35	2.37	2.38	2.38	2.39

Bolded values were tested. All other values are simulated and for reference only.